

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

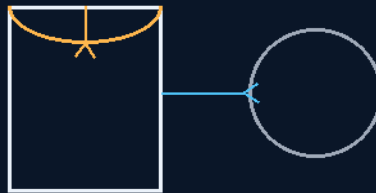
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 162

PLUNGER-LID LAUNDRY SYSTEM AND GREENHOUSE DRYING LOOP

The loader that is the lid
the machine died but the containers lived



superseded, never deleted

PLUNGER LID · DROP POD · DRY ROOM

CONTAINERS OUTLIVE MACHINES

Sanitation system — v1.0 in force — MACHINE OBSOLETE

GP-IM-162

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — MECHANICAL PASS · MACHINE SUPERSEDED

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 162

PHASE 162: PLUNGER-LID LAUNDRY SYSTEM AND GREENHOUSE DRYING LOOP — STATUS: CURRENT WITH OBSOLETE ELEMENT — THE WASHING-MACHINE DESCRIPTION SUPERSEDED BY THE PHASE 191 ROLLER-MANGLE LINE (AUTHOR'S RULING, SEATED); THE BASKET-AND-PLUNGER ARCHITECTURE LIVES — IT SEEDS PHASE 167; MECHANICAL ARCHITECTURE PASS (VET BATCH 20). The machine died but the containers lived — his words, and the phase's fate. Phase 162's heart survives untouched: the plunger-lid basket, a top-load cylinder whose flexible mesh lid inverts into a twist-lock plunger, sealing the load inside its own basket — the loader that is the lid. Around it stood a sealed-drum wash, an air-swept spin driving water to wall guttering, and a greenhouse drying room; on the author's ruling the washing-machine description is marked OBSOLETE — superseded by the Phase 191 roller-mangle line (pressure wash through the clothes, no spin, the pizza-oven profile) — and the obsolete text stands untouched in §1 per doctrine, never silently deleted. The basket's route is now the laundry room's large drop pod. What the phase invented outlives its machine: the basket-and-plunger architecture is the parent of Phase 167's universal container. The vet graded the mechanical architecture a pass.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-162, v1.0 — 23 August 2026. The one hundred and sixty-third manual of the program — 163 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the laundry law as seated (contents line, the bodies — the plunger-lid basket, the sealed drum, the air-swept spin, the greenhouse drying room — the physics note, the provenance, and the OBSOLETE ruling on the machine description, all verbatim, obsolete text untouched per doctrine) — §2 why the plunger matters (graded) — §3 the system in the drum — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 162: **Plunger-Lid Laundry System & Greenhouse Drying Loop.** Top-load cylinder basket with flexible mesh lid that inverts into a twist-lock plunger, sealing inside the basket; air-swept spin drives water to wall guttering without a second outer bowl; extractor-level hot drying room vents humidity to the greenhouses.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Full-Crew Laundry Capability Using the Fleet's Existing Water, Gutter, and Heat Loops'

LIVE-INVENTED (29 July 2026): the first phase of the post-161 era — designed in conversation the day the bathroom suite closed. Phase number assigned provisionally; final numbering reserved for Archer.

Crew Systems Baseline: Zero-G Laundry Without Front-Load or Top-Load Failure Modes

THE PLUNGER-LID BASKET (THE LOADER THAT IS THE LID) (VERBATIM)

- **The Basket:** a cylinder loaded at bench height — clothes in one end, a flexible mesh circular cover flexed into place over them. No front door to float clothes out of, no top tub to stuff down into.
- **The Mount:** at the laundry, remove the mesh cover, place the basket upside down on the matching machine opening, grab the molded-in side handles, and twist-lock — the basket locks and seals to the machine at the rim.
- **The Plunger:** what was the basket's base is now the plunger — hand on the top plunger handle, push down along matching grooves in the basket's inner walls to maximum lid level, and twist-lock to the inside of the basket. The plunger seals the basket's interior, so the wash water has nowhere to escape upward. The load stays captured inside the basket, compressed to working level; the part you fill is the part that seals, and the base you loaded against is the ram that holds it.

THE SEALED DRUM & AGITATION WASH (VERBATIM)

- **Drum and Actuators:** standard drum mechanics work perfectly in space — the unit is sealed, so air-fluid dynamics work in the fleet's favour.
- **Water Level:** at least half full — agitation moves the water through the plunger-held load exactly as a terrestrial machine, with the plunger keeping clothes in the water column instead of floating free.
- **Water Budget Reality:** plants use more water than a crew of a hundred ever will for washing — laundry rides the agriculture surplus; wet washing is not the constraint.

THE AIR-SWEPT SPIN CYCLE (VERBATIM)

- **Single-Wall Extraction:** in spin mode, the air trapped inside the sealed drum sweeps the water into the outer wall gutter — no second spinning outer bowl required. The spinner drains to the same wall guttering as the sanitation suite.
- **Shared Drainage:** clothing water goes where shower water goes — one reclamation path for the whole module.

THE GREENHOUSE DRYING ROOM (VERBATIM)

- **Extractor-Level Spin:** the machine spins to extractor dryness before anything leaves it.
- **The Hot Room:** a warm room with air blowing downward over hanging clothes — the downdraft forces the clothes to hang vertically and dries them without a tumble drum.
- **Humidity Harvest:** the room's humid exhaust vents straight to the greenhouses — the crew's laundry waters the farm's air.

PHYSICS NOTE — PHASE 162 (KIMI AUDIT: AFFIRMED) (VERBATIM)

- **Compression washing is proven physics.** Plunger washers have cleaned clothes by hand for a century — water driven through fabric under pressure and release. The basket-lid-plunger turns that into the zero-G loader: clothes are captured before they can float, compressed to the working level, and washed by ordinary agitation.
- **The air-swept spin is the toilet family again.** A sealed spinning drum co-rotates its trapped air; water, being denser, migrates outward through the air to the wall — the same law as the Phase 137 bowl and its 3-inch gutter. Deleting the second outer bowl is correct simplification: the rim gutter already exists elsewhere in the suite and catches fling-off directly.
- **The drying room closes two loops at once.** Warm downdraft orients hanging clothes by drag (they align with the flow — no clips, no floating), and the moisture it strips is plant food: greenhouses are the ship's humidity sinks, so every gram of laundry water ends up in the farm's atmosphere. Mass is neither lost nor pumped uphill.

ORIGIN OF THOUGHT — PHASE 162 (VERBATIM)

Archer, 29 July 2026: "front loaders are out, top loaders will be clumsy stuffing clothes. so the laundry load level basket is the lid... the basket is a cylinder the clothes go in one end with a flexible mesh circular cover you flex into place, over them, when full take to laundry, remove cover, place upside down on the matching opening, grab the molded in side handles and twist lock the seal. place hand on top to the plunger handle push down, to maximum lid level twist and lock, what was the base is your plunger, set into matching grooved sides for smooth operation. machine will wash as usual as the water

will move by agitation, yes it must be at least half full, but plants use a lot of water, more than a crew of a hundred would wash. in the spin mode the air trapped inside will sweep the water into the outer wall gutter without the need for a second spinning bowl outer. not too much else to alter, clothing water goes where shower water goes. the machine will operate to extractor level, a hot room with warm air blowing down for hanging clothes vented to humidity for the greenhouses and additionally will force the clothes to hang vertically."

Why it matters, in plain English: a hundred people on a century ship generate a mountain of laundry, and every terrestrial machine assumes gravity in its door. Archer's answer follows the fleet's oldest rule — the part you fill is the part that seals — then goes one better: the base of the loading basket comes back as the wash plunger, so the loader, the lid, and the ram are one piece of hardware with three jobs. Nothing about the machine fights space; the spin borrows the toilet's wall-gutter law, the drains borrow the shower's, and the drying room pays its moisture forward to the farms. Laundry day on a starship ends with tomato plants breathing what the socks gave off.

Amendment — 30 July 2026, Archer, verbatim (routing change, machine obsolescence recorded):

"of note we need to mark the carry drum laundry basket as now not straight to machine but to laundry room large drop pod, as that washing machine description was made obsolete, be we gained the containers from it"

Recorded as instructed (KIMI): the carry drum laundry basket no longer docks straight to a machine — its route is now to the laundry room's large drop pod. The washing machine description this phase carried is hereby marked OBSOLETE, superseded by the Phase 191 roller-mangle line (pressure wash through the clothes, no spin, the pizza-oven profile); the obsolete text stands untouched below per doctrine — superseded, never silently deleted. And his closing clause is the engineering truth of the whole sitting: the machine died but the containers lived — this phase's basket-and-plunger architecture is exactly what became the Phase 167 universal plunger container, the fleet's 10L/10kg supply-chain standard. Nothing was wasted; the laundry machine was the container program's parent. Flow as now amended: carry drum basket -> laundry room large drop pod -> Phase 191 line. Flag, kept honest and held for Archer (not designed): whether the drop pod meters loads straight into the 191 rollers or holds batch for a crew member to feed — his call when the laundry room layout gets its own pass.

Amendment — 31 July 2026, Archer, verbatim (the held drop-pod question ANSWERED — location, dual function, and the plunger transfer):

"laundry drop pod is located in the deck floor above the laundry, yes you could load sideways, but in gravity no, must be dual function capable, and the height of a multiroom collection pod/container (not pod as in a ship pod) like a big round tank, to high to hand load without a ladder; is fed the same way we were going to feed the original washing machine, except the main pod has a top door seal of its own,

place basket lock in place, main laundry container door opens , open basket door, plunger pushed, basket door closed , main door closes, plunger up, return to room."

Audit acknowledgment (KIMI): the held question — meter straight into the line or hold batch — is answered with a third architecture that is better than either: the drop pod is a multiroom COLLECTION tank in the deck floor above the laundry (not a ship pod — his disambiguation verbatim), tall as a big round tank, too high to hand-load without a ladder, dual-function by law: sideways loading is possible in zero-G, but in gravity the load must drop — so it feeds the way the original Phase 162 machine was going to be fed, the architecture that survived the machine's own obsolescence. The transfer sequence is now law: place basket, lock in place, main laundry container door opens (the pod has a top door seal of its own), open basket door, plunger pushed, basket door closed, main door closes, plunger up, return to room. Every move in that chain is gravity-indifferent — the plunger does in zero-G what falling does in a gravity field — which is exactly what dual function demands. The plunger-lid lineage is complete: Phase 162 birthed it, Phase 167 made it the supply chain, and now it closes the loop as the laundry's own throat. Flag, kept honest: the ladder note is recorded as written — access to the pod rim for maintenance is a layout detail for the laundry room pass, his call when that pass comes.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PLUNGER-LID, AFFIRMED (the phase's own physics note): a lid that inverts into a plunger makes the container its own machine — seal, agitate, and empty with one part and no gravity. It is the same gravity-abolishing move the audit affirmed at Phase 167: pressure, not pouring.

THE OBSOLESCENCE RULING, SEATED STRAIGHT: the roller-mangle line (Phase 191) washes by pressure through the clothes — no spin cycle, simpler machine, the pizza-oven profile — so the drum-and-spin description retires, and the file keeps it whole: superseded, never deleted. The basket docks now to the drop pod, and the phase's engineering truth stands as he closed it: the machine died but the containers lived.

THE AIR-SWEPT AND GREENHOUSE BODIES: seated with the phase; their loop logic (air does the carrying, the room does the drying) is sound household physics and rides forward into the pod-lineage machines.

§3 — THE SYSTEM IN THE DRUM

- THE BASKET: top-load cylinder; flexible mesh lid inverting to a twist-lock plunger — the loader that is the lid.
- THE ROUTE: basket to the laundry room's large drop pod (his ruling — no straight-to-machine dock).

- THE OBSOLETE BODY: the sealed-drum wash and air-swept spin — superseded by Phase 191, preserved untouched in §1.
- THE DRYING: the greenhouse drying room — the loop's warm end.
- THE LEGACY: the basket-and-plunger architecture — parent of Phase 167's universal container.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE LOADER-IS-THE-LID DOCTRINE (seated with the phase): one part does the whole handling chain — seal, plunge, empty.
- THE SUPERSESSION DOCTRINE (his ruling): obsolete means superseded, never deleted — the obsolete text stands untouched, marked.
- THE CONTAINERS-OUTLIVE-MACHINES DOCTRINE: his closing clause, the phase's epitaph and its legacy — 167 is the proof.
- THE GRAVITY-ABOLISHING FAMILY: plunger, pod, mangle — the laundry line washes by pressure in any gravity.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 20 (15 August 2026 — phases 161-175, cross-checked in sitting 307), SEATED. The grade, verbatim from the batch: '162 → PASS.' The ledger line: '162 mechanical architecture PASS.' No red, no yellow on the mechanics. THE OBSOLESCENCE is the author's own later ruling — the washing-machine description superseded by the Phase 191 roller-mangle line; the basket route amended to the drop pod; both verbatim in §1, the obsolete text preserved per doctrine.

§6 — BUILD SEQUENCE

- STEP 1 — Build the basket: mesh lid, twist-lock plunger, seal integrity logged.
- STEP 2 — Prove the plunger: inversion cycles, seal life, emptying completeness in 1G and parabolic zero-G.
- STEP 3 — Dock the route: basket-to-drop-pod transfer drill (the machine dock is retired — his ruling).
- STEP 4 — Run the drying room: greenhouse loop duty cycle.
- STEP 5 — Hand the architecture forward: the Phase 167 container derivatives from this basket.
- STEP 6 — Publish the ledger: seal data, cycle counts, the supersession record — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 191 — the roller-mangle line: the superseding machine (pressure wash, no spin).
- PHASE 167 — the universal plunger container (GP-IM-167): this basket's child.
- PHASE 159 — the agricultural matrix (GP-IM-159): the greenhouse air family.
- PHASE 137 — the waste core (GP-IM-137): the centrifugal gutter physics cousins.

§8 — DO-NOT-BUILD

- DO NOT build the drum machine — OBSOLETE by his ruling; the Phase 191 roller-mangle line is the washer.
- DO NOT delete the obsolete text — superseded, never silently deleted (doctrine, his law).
- DO NOT dock the basket to a machine — the route is the drop pod, seated.
- DO NOT lose the plunger — the lid-that-plunges is the phase's living invention; 167 runs on it.

§9 — OWED

OWED: nothing flagged by the vet — a clean mechanical pass. Standing duty: the basket's seal-life and cycle data publish from the bench (§6); the supersession record is seated and permanent. The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-162, v1.0 — CURRENT WITH OBSOLETE ELEMENT — the washing-machine description marked OBSOLETE on the author's ruling (superseded by the Phase 191 roller-mangle line; the obsolete text stands untouched in §1 per doctrine); the basket-and-plunger architecture lives — it seeds Phase 167, seated law, master v2.1 (numerical print order). The one hundred and sixty-third manual of the program — 163 of 290. Built 23 August 2026, fifteenth of the 20-manual 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'. The

phase's own chain, all seated in the master: the contents line and the bodies (as seated); THE OBSOLESCENCE RULING — the washing-machine description marked OBSOLETE on the author's word, superseded by the Phase 191 roller-mangle line, the obsolete text standing untouched in §1 per doctrine; the basket route amended to the drop pod; the vet batch 20 mechanical pass in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the basket bench data, or his word, or his word.

END OF MANUAL — PHASE 162